SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA5117
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO5: *Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.CO* (BL5, BL6)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **TOTAL MARKS: 50**

Annexure A
CONSTRAINT-BASED PROBLEM SOLVING

Code of Conduct:

I Harshini R N (192421175) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Harshini R N

Annexure A CONSTRAINT-BASED PROBLEM SOLVING

1. A financial institution must secure all web transactions using TLS, but legacy systems only support older SSL versions. Design a solution to ensure secure communication.

   Constraints:

A. Backward compatibility must be maintained for 20% legacy clients
B. Sensitive data must not be transmitted using SSL
C. System downtime must be minimal during migration
D. Cost of implementation should be optimized

2. An e-commerce platform is considering implementing the SET protocol for payment security but currently uses TLS-based encryption. Analyze and propose a decision.

   Constraints:

A. Must support high transaction throughput (10,000 transactions/minute)
B. Compatibility with modern browsers is required
C. Implementation complexity should be minimized
D. Regulatory compliance (PCI-DSS) must be ensured

3. A corporate network is experiencing repeated malware infections that spread rapidly without user interaction. Propose a containment and prevention strategy.

   Constraints:

A. The network consists of 500+ interconnected systems
B. Downtime per system should not exceed 10 minutes
C. Detection must occur within 2 minutes of infection
D. Solution must differentiate between virus, worm, and Trojan behavior

4. An organization needs to deploy a firewall architecture to protect its internal network from external threats and insider attacks. Design the solution.

   Constraints:

A. Must support both packet filtering and application-level inspection
B. Latency introduced should not exceed 5 ms per request
C. Should handle at least 1 Gbps traffic
D. Must allow secure remote access for employees

5. A large enterprise network generates excessive false positives in its IDS, leading to alert fatigue. Propose an optimized IDS/IPS solution.

Constraints:

A. Detection accuracy must be at least 95%
B. False positive rate should be reduced below 5%
C. System must process real-time traffic of 2 Gbps
D. Solution should integrate with existing firewall infrastructure

ANSWERS:

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1. Financial Institution – TLS Migration with Legacy SSL

SOLUTION:

- Deploy a dual-protocol gateway (TLS + SSL termination proxy).
- Configure TLS as the default for all modern clients.
- For 20% legacy clients, allow SSL connection only to a proxy layer.
- Proxy immediately re-encrypts SSL traffic into TLS before reaching core servers.
- Sensitive data never transmitted over SSL directly; proxy enforces TLS internally.
- Gradual migration plan: upgrade legacy clients over time.
- Minimal downtime: phased rollout using load balancers.
- Cost optimized: reuse existing infrastructure with reverse proxy and TLS offloading.

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2. E-Commerce Platform – SET vs TLS Decision

ANALYSIS:

- SET protocol: strong security but complex, requires special client software, low browser support.
- TLS: widely supported, efficient, integrated with modern browsers, PCI-DSS compliant.

DECISION:

- Continue with TLS-based encryption.
- Justification:

  - TLS supports 10,000+ transactions/minute throughput.
  - Compatible with all modern browsers.
  - Lower implementation complexity compared to SET.
  - Meets PCI-DSS regulatory compliance.
- SET rejected due to complexity, poor adoption, and incompatibility.

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3. Corporate Network – Malware Containment & Prevention

STRATEGY:

- Deploy enterprise-grade IDS/IPS for rapid detection (<2 minutes).
- Use network segmentation to isolate infected systems quickly.
- Automated quarantine: disconnect infected nodes within 10 minutes.
- Apply patches and endpoint protection across all 500+ systems.
- Differentiate malware:
  - Virus → file-based infection.
  - Worm → self-replicating, no user interaction.
  - Trojan → disguised software requiring user action.
- Implement centralized logging and behavioral analysis to classify threats.
- Educate users to avoid Trojan installations.

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4. Firewall Architecture – Internal & External Protection

DESIGN:

- Deploy a Next-Generation Firewall (NGFW).
- Features:
  - Packet filtering for IP/port/protocol.
  - Application-level inspection for deep packet analysis.
  - Latency <5 ms per request ensured via hardware acceleration.
  - Throughput capacity ≥1 Gbps.
- Architecture:

  - Perimeter firewall for external threats.

  - Internal segmentation firewall for insider threats.

  - VPN gateway integrated for secure remote employee access.

- Policy:

  - Default deny, allow only authorized traffic.

  - Role-based access for employees.

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5. IDS/IPS Optimization – Reducing False Positives

SOLUTION:

- Deploy hybrid IDS/IPS with machine learning anomaly detection + signature-based rules.
- Tune detection thresholds to balance sensitivity and accuracy.
- Regularly update signature databases.
- Implement feedback loop: false positives reviewed and rules adjusted.
- Integrate IDS/IPS with firewall for coordinated response.
- Performance:
  - Real-time traffic handling ≥2 Gbps.
  - Detection accuracy ≥95%.
  - False positive rate <5%.
- Dashboard for SOC team to reduce alert fatigue.

SUMMARY:

- TLS migration: dual-protocol proxy ensures backward compatibility.
- E-commerce: TLS preferred over SET for throughput, compatibility, compliance.
- Malware containment: IDS/IPS + segmentation + rapid quarantine.
- Firewall: NGFW with packet + application inspection, VPN for remote access.
- IDS/IPS optimization: hybrid detection, ML tuning, firewall integration.